# Solution Sheet 3— Stochastic Calculus

**3.1:** (a) By Itô's Lemma, if $Y_t = \mathrm{e}^{\lambda t} X_t$, then

$$\begin{aligned}\mathrm{d}Y_t &= \mathrm{e}^{\lambda t}\lambda(\mu - X_t) + \mathrm{e}^{\lambda t}\sigma\,\mathrm{d}W_t + \lambda \mathrm{e}^{\lambda t} X_t\,\mathrm{d}t \\ &= \lambda\mu\mathrm{e}^{\lambda t}\,\mathrm{d}t + \sigma\mathrm{e}^{\lambda t}\,\mathrm{d}W_t,\end{aligned}$$

so

$$Y_t = X_0 + \mu(\mathrm{e}^{\lambda t} - 1) + \sigma\int_0^t \mathrm{e}^{\lambda s}\,\mathrm{d}W_s,$$

and

$$X_t = \mathrm{e}^{\lambda t}X_0 + \mu(1 - \mathrm{e}^{-\lambda t}) + \sigma\mathrm{e}^{-\lambda t}\int_0^t \mathrm{e}^{\lambda s}\,\mathrm{d}W_s.$$

(b) We find the generating function of $Y$ by considering $Z_t = \mathrm{e}^{\eta Y_t}$. By Itô's Lemma,

$$\begin{aligned}\mathrm{d}Z_t &= \eta Z_t\,\mathrm{d}Y_t + \frac{1}{2}\eta^2 Z_t\,\mathrm{d}\langle Y\rangle_t \\ &= \left(\eta Z_t\lambda\mu\mathrm{e}^{\lambda t} + \frac{1}{2}\eta^2 Z_t\sigma^2\mathrm{e}^{2\lambda t}\right)\mathrm{d}t + \eta Z_t\sigma\mathrm{e}^{\lambda t}\,\mathrm{d}W_t.\end{aligned}$$

Then, provided $\mathbb{E}\left[\int_0^t Z_s^2\mathrm{e}^{2\lambda s}\mathrm{d}s\right] < \infty$,

$$\begin{aligned}m(t) = \mathbb{E}\left[Z_t\right] &= \int_0^t \left(\eta\lambda\mu\mathrm{e}^{\lambda s} + \frac{1}{2}\eta^2\sigma^2\mathrm{e}^{2\lambda s}\right) m(s)\mathrm{d}s \\ \implies \dot{m}(t) &= \left(\eta\lambda\mu\mathrm{e}^{\lambda t} + \frac{1}{2}\eta^2\sigma^2\mathrm{e}^{2\lambda t}\right) m(t) \\ \implies m(t) &= A\exp\left(\eta\mu(\mathrm{e}^{\lambda t} - 1) + \frac{1}{2}\eta^2\frac{\sigma^2}{2\lambda}(\mathrm{e}^{2\lambda t} - 1)\right) \\ &= \exp\left(\eta(x_0 + \mu(\mathrm{e}^{\lambda t} - 1)) + \frac{1}{2}\eta^2\frac{\sigma^2}{2\lambda}(\mathrm{e}^{2\lambda t} - 1)\right),\end{aligned}$$

since $m(0) = \mathbb{E}\left[\mathrm{e}^{\eta x_0}\right]$. Then, we can see from the form of the generating function that

$$\begin{aligned}Y_t &\sim N\left(x_0 + \mu(\mathrm{e}^{\lambda t} - 1), \frac{\sigma^2}{2\lambda}(\mathrm{e}^{2\lambda t} - 1)\right) \\ \implies \mathrm{e}^{-\lambda t}Y_t = X_t &\sim N\left(\mathrm{e}^{-\lambda t}x_0 + \mu(1 - \mathrm{e}^{-\lambda t}), \frac{\sigma^2}{2\lambda}(1 - \mathrm{e}^{-2\lambda t})\right).\end{aligned}$$

(c) We still have

$$m(t) = A\exp\left(\eta\mu(\mathrm{e}^{\lambda t} - 1) + \frac{1}{2}\eta^2\frac{\sigma^2}{2\lambda}(\mathrm{e}^{2\lambda t} - 1)\right),$$

but now $Y_0 \sim N\left(\mu, \frac{\sigma^2}{2\lambda}\right)$ and so $m(0) = \mathbb{E}\left[\mathrm{e}^{\eta Y_0}\right] = \mathrm{e}^{\eta\mu + \frac{1}{2}\eta^2\frac{\sigma^2}{2\lambda}}$. Therefore,

$$\begin{aligned} m(t) &= \exp\left(\eta\mu\mathrm{e}^{\lambda t} + \frac{1}{2}\eta^2\frac{\sigma^2}{2\lambda}\mathrm{e}^{2\lambda t}\right) \\ \implies Y_t &\sim N\left(\mu\mathrm{e}^{\lambda t}, \frac{\sigma^2}{2\lambda}\mathrm{e}^{2\lambda t}\right) \\ \implies X_t &\sim N\left(\mu, \frac{\sigma^2}{2\lambda}\right), \end{aligned}$$

so $X$ is stationary.

**3.2:** We work via induction, and we have shown the result for $n = 1, 2, 3$ in the previous sheet, so the base case is done. Suppose the result holds for some $n$. Then,

$$\int_{u_{n+1}=0}^{t}\int_{u_n=0}^{u_{n+1}}\cdots\int_{u_1=0}^{u_1}\mathrm{d}W_{u_1}\cdots\mathrm{d}W_{u_n}\mathrm{d}W_{u_{n+1}} = \int_0^t \frac{u^{\frac{n}{2}}}{n!}h_n\left(\frac{W_u}{\sqrt{u}}\right)\mathrm{d}W_u.$$

Now, by Itô's Lemma, if we let $Z_t = \frac{t^{\frac{n+1}{2}}}{(n+1)!}h_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)$, then

$$\begin{aligned} \mathrm{d}Z_t &= \frac{1}{2}\frac{t^{\frac{n-1}{2}}}{n!}h_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}t - \frac{1}{2}\frac{t^{\frac{n-2}{2}}}{(n+1)!}W_t h'_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}t \\ &\qquad + \frac{t^{\frac{n}{2}}}{(n+1)!}h'_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}W_t + \frac{1}{2}\frac{t^{\frac{n-1}{2}}}{(n+1)!}h''_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}t \\ &= \frac{1}{2}\frac{t^{\frac{n-1}{2}}}{(n+1)!}\left[(n+1)h_{n+1}\left(\frac{W_t}{\sqrt{t}}\right) - \frac{W_t}{\sqrt{t}}h'_{n+1}\left(\frac{W_t}{\sqrt{t}}\right) + h''_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\right]\mathrm{d}t \\ &\qquad + \frac{t^{\frac{n}{2}}}{n!}h_n\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}W_t \\ &= \frac{1}{2}\frac{t^{\frac{n-1}{2}}}{(n+1)!}\left[(n+1)h_{n+1}\left(\frac{W_t}{\sqrt{t}}\right) - (n+1)\frac{W_t}{\sqrt{t}}h_n\left(\frac{W_t}{\sqrt{t}}\right) + n(n+1)h''_{n+1}\left(\frac{W_t}{\sqrt{t}}\right)\right]\mathrm{d}t \\ &\qquad + \frac{t^{\frac{n}{2}}}{n!}h_n\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}W_t \\ &= \frac{t^{\frac{n}{2}}}{n!}h_n\left(\frac{W_t}{\sqrt{t}}\right)\mathrm{d}W_t, \end{aligned}$$

using the given relations for Hermite polynomials. So then

$$Z_t = \frac{t^{\frac{n+1}{2}}}{(n+1)!}h_{n+1}\left(\frac{W_t}{\sqrt{t}}\right) = \int_0^t \frac{u^{\frac{n}{2}}}{n!}h_n\left(\frac{W_u}{\sqrt{u}}\right)\mathrm{d}W_u,$$

which is exactly what we had above. Therefore, by induction, our result holds for all $n$.

**3.3:** (a) Let $X_t = (Y_t^1)^2 + (Y_t^2)^2$. Then, by Itô's Lemma,

$$\begin{aligned}
\mathrm{d}X_t &= 2Y_t^1\,\mathrm{d}Y_t^2 + 2Y_t^2\,\mathrm{d}Y_t^1 + \frac{1}{2}2\,\mathrm{d}\langle Y_t^2\rangle + \frac{1}{2}2\,\mathrm{d}\langle Y_t^1\rangle \\
&= 2Y_t^1\left(-\frac{1}{2}Y_t^2\,\mathrm{d}t + Y_t^1\,\mathrm{d}W_t\right) + 2Y_t^2\left(-\frac{1}{2}Y_t^1\,\mathrm{d}t - Y_t^2\,\mathrm{d}W_t\right) \\
&\qquad + \left(Y_t^1\right)^2\,\mathrm{d}t + \left(Y_t^2\right)^2\,\mathrm{d}t \\
&= \left[-\left(Y_t^1\right)^2 + \left(Y_t^2\right)^2 - \left(Y_t^2\right)^2 + \left(Y_t^1\right)^2\right]\mathrm{d}t + \left[-2Y_t^1Y_t^2 + 2Y_t^1Y^2 - t\right]\mathrm{d}W_t \\
&= 0,
\end{aligned}$$

and so $X_t$ is constant.

(b) Again by Itô's Lemma,

$$\begin{aligned}
\mathrm{d}(\cos(Z_t)) &= -\sin(Z_t)\,\mathrm{d}Z_t - \frac{1}{2}\cos(Z_t)\,\mathrm{d}t \\
\mathrm{d}(\sin(Z_t)) &= \cos(Z_t)\,\mathrm{d}Z_t - \frac{1}{2}\sin(Z_t)\,\mathrm{d}t,
\end{aligned}$$

so we can see that $Z_t = W_t$, a standard Brownian motion, will do.

(c) We can see that our process $\boldsymbol{Y}_t$ lies on the unit circle and its angle with the $x$-axis is given by $Z_t$, now a Brownian motion. If $N_t$ is the winding number then we can write $N_t = \left\lfloor \frac{Z_t}{2\pi} \right\rfloor$, so

$$\begin{aligned}
\mathbb{P}\left(N_t \geq n\right) &= \mathbb{P}\left(\left\lfloor \frac{Z_t}{2\pi} \right\rfloor \geq n\right) \\
&= \mathbb{P}\left(\frac{Z_t}{2\pi} \geq n\right) \\
&= \mathbb{P}\left(Z_t \geq 2n\pi\right) \\
&= \Phi\left(\frac{-2n\pi}{\sqrt{t}}\right),
\end{aligned}$$

since $Z_t \sim N(0,t)$, where $\Phi$ is the c.d.f. of a standard normal r.v..

**3.4:** From the differential form we see that

$$X_t = x_0 + \int_0^t \mu X_s\,\mathrm{d}s + \int_0^t \sigma X_s\,\mathrm{d}W_s,$$

and so

$$m(t) = \mathbb{E}\left[X_t\right] = x_0 + \mu\int_0^t m(s)\,\mathrm{d}s,$$

provided that we have $\mathbb{E}\left[\sigma^2\int_0^t X_s^2\,\mathrm{d}s\right] < \infty$ (which can be shown). But then, differentiating, we get

$$\begin{aligned}
\dot{m}(t) &= \mu m(t) \\
\implies m(t) &= A\exp(\mu t) \\
\implies m(t) &= x_0\exp(\mu t).
\end{aligned}$$

Taking limits gives

$$\lim_{t\to\infty} \mathbb{E}\left[X_t\right] = \infty, \quad \text{for } \alpha > 0.$$

We know, or it can be verified easily by Itô's Lemma, that

$$\begin{aligned} X_t &= x_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right) \\ &= x_0 \exp\left(\sigma\left(W_t - \frac{1}{\sigma}\left(\frac{1}{2}\sigma^2 - \mu\right)t\right)\right) \\ &= x_0 \exp\left(\sigma\left(W_t - at\right)\right), \end{aligned}$$

for $a = \frac{1}{\sigma}\left(\frac{1}{2}\sigma^2 - \mu\right) > 0$ when $0 < \mu < \frac{\sigma^2}{2}$, so then we can see that

$$\lim_{t\to\infty} X_t = 0.$$

**3.5:** We can show that the given $X_t$ satisfies the SDE, and then argue by uniqueness of solutions. By Itô's Lemma,

$$\begin{aligned} \mathrm{d}X_t &= \left(y - x - \int_0^t \frac{\mathrm{d}W_s}{1-s}\right) \mathrm{d}t + \mathrm{d}W_t \\ &= \frac{1}{1-t}\left(y - x - (y-x)t - (1-t)\int_0^t \frac{\mathrm{d}W_s}{1-s}\right) \mathrm{d}t + \mathrm{d}W_t \\ &= \frac{y - X_t}{1-t}\,\mathrm{d}t + \mathrm{d}W_t, \end{aligned}$$

so the chosen $X$ satisfies our SDE. We can argue using Theorem 5.5 that this must be the unique solution. To write our SDE in the form used in the theorem, we have $\mu(t,z) = \frac{y-z}{1-t}$, $\sigma(t,z) = 1$ and we suppose $t \in [0,T]$ where $T < 1$, so

$$\begin{aligned} |\mu(t,z)| + |\sigma(t,z)| &= 1 + \frac{|y-z|}{1-t} \\ &\leq 1 + \frac{|y|}{1-t} + \frac{|z|}{1-t} \\ &\leq \frac{1+|y|}{1-t}(1+|z|) \\ &\leq \frac{1+|y|}{1-T}(1+|z|). \end{aligned}$$

Similarly,

$$\begin{aligned} |\mu(t,z) + \mu(t,\tilde{z})| + |\sigma(t,z) + \sigma(t,\tilde{z})| &= \frac{|y - z - (y - \tilde{z})|}{1-t} \\ &= \frac{|z - \tilde{z}|}{1-t} \\ &\leq \frac{|z-\tilde{z}|}{1-T}, \end{aligned}$$

and so we satisfy the conditions required in Theorem 5.5. The theorem then tells us that our SDE has a unique solution for each $T < 1$, and so this unique solution must be $X$ given as above. Moreover, this is independent of $T$, so we can define this to be the solution for all $t < 1$. Now, we have that since $\mathbb{E}\left[\int_0^t \frac{\mathrm{d}s}{(1-s)^2}\right] < \infty$, our stochastic integral term is a martingale, and in particular, $\mathbb{E}\left[\int_0^t \frac{\mathrm{d}W_s}{1-s}\right] = 0$. Then we find that, by the Itô Isometry,

$$\begin{aligned}
\mathbb{E}\left[(X_t - y)^2\right] &= (1-t)^2(x-y)^2 + (1-t)^2\mathbb{E}\left[\int_0^t \frac{\mathrm{d}s}{(1-s)^2}\right] \\
&= (1-t)^2(x-y)^2 + (1-t)^2\left(\frac{1}{1-t} - 1\right) \\
&\to 0 \quad \text{as } t \to 1,
\end{aligned}$$

and so $X_t \to y$ in $\mathcal{L}^2(\mathbb{P})$ as $t \to 1$.

**3.6:** We have $F_t = \exp\left\{-\alpha B_t + \frac{1}{2}\alpha^2 t\right\}$, so $F_t = f(t, B_t)$ where $f(t,x) = \mathrm{e}^{-\alpha x + \alpha^2 t/2}$. Then

$$\begin{aligned}
\frac{\partial f}{\partial x} &= -\alpha f(t,x) \\
\frac{\partial^2 f}{\partial x^2} &= \alpha^2 f(t,x) \\
\frac{\partial f}{\partial t} &= \frac{1}{2}\alpha^2 f(t,x)
\end{aligned}$$

Hence

$$\begin{aligned}
\mathrm{d}f(t,B_t) &= -\alpha f(t,B_t)\,\mathrm{d}B_t + \left(\frac{1}{2}\alpha^2 f(t,B_t) + \frac{1}{2}\alpha^2 f(t,B_t)\right)\mathrm{d}t \\
&= -\alpha F_t\,\mathrm{d}B_t + \alpha^2 F_t\,\mathrm{d}t.
\end{aligned}$$

Then

$$\begin{aligned}
\mathrm{d}(X_t F_t) &= F_t\,\mathrm{d}X_t + X_t\,\mathrm{d}F_t + \mathrm{d}F_t\,\mathrm{d}X_t \\
&= F_t \alpha X_t\,\mathrm{d}B_t + F_t r\,\mathrm{d}t - \alpha F_t X_t\,\mathrm{d}B_t + \alpha^2 F_t X_t\,\mathrm{d}t \\
&\qquad + \left(-\alpha F_t\,\mathrm{d}B_t + \alpha^2 F_t\,\mathrm{d}t\right)(\alpha X_t\,\mathrm{d}B_t + r\,\mathrm{d}t) \\
&= (\alpha F_t X_t - \alpha F_t X_t)\,\mathrm{d}B_t + \left(rF_t + \alpha^2 F_t X_t - \alpha^2 F_t X_t\right)\mathrm{d}t \\
&= rF_t\,\mathrm{d}t.
\end{aligned}$$

Here we used $(\mathrm{d}t)^2 = \mathrm{d}B_t\,\mathrm{d}t = 0$ between the second and third lines.

In particular, we have shown that $\mathrm{d}Y_t = r\exp\left\{-\alpha B_t + \frac{1}{2}\alpha^2 t\right\}\,\mathrm{d}t$.

It follows that

$$\begin{aligned}
&\int_0^t \mathrm{d}(X_s F_s) = F_t X_t - X_0 F_0 \\
\implies\quad & F_t X_t - X_0 F_0 = \int_0^t rF_s\,\mathrm{d}s \\
&\qquad = r\int_0^t \exp\left\{-\alpha B_s + \frac{1}{2}\alpha^2 s\right\}\mathrm{d}s \\
\implies\quad & X_t \mathrm{e}^{-\alpha B_t + \alpha^2 t/2} = x_0 \mathrm{e}^0 + r\int_0^t \exp\left\{-\alpha B_s + \frac{1}{2}\alpha^2 s\right\}\mathrm{d}s \\
\implies\quad & X_t = x_0 \mathrm{e}^{\alpha B_t - \alpha^2 t/2} + r\int_0^t \exp\left\{\alpha(B_t - B_s) - \frac{1}{2}\alpha^2(t-s)\right\}\mathrm{d}s.
\end{aligned}$$

**3.7:** We wish to use Itô's Lemma on $R_t = f(W_t^1, \ldots, W_t^n)$ with $f(x_1, \ldots, x_n) = (x_1^2 + \cdots + x_n^2)^{\frac{1}{2}}$, so

$$\begin{aligned}\frac{\partial f}{\partial x_i} &= \frac{x_i}{(x_1^2 + \cdots + x_n^2)^{\frac{1}{2}}}\\ \frac{\partial^2 f}{\partial x_i^2} &= \frac{1}{(x_1^2 + \cdots + x_n^2)^{\frac{1}{2}}}\left(1 - \frac{x_i^2}{(x_1^2 + \cdots + x_n^2)^{\frac{3}{2}}}\right),\end{aligned}$$

and then,

$$\begin{aligned}\mathrm{d}R_t &= \sum_{i=1}^n \frac{W_t^i \,\mathrm{d}W_t^i}{R_t} + \sum_{i=1}^n \frac{1}{2}\frac{1}{R_t}\left(1 - \frac{(W_t^i)^2}{\sum_i (W_t^i)^2}\right)\mathrm{d}t\\ &= \sum_{i=1}^n \frac{W_t^i \,\mathrm{d}W_t^i}{R_t} + \frac{1}{2}\frac{1}{R_t}\left(1 - \frac{\sum_i (W_t^i)^2}{\sum_i (W_t^i)^2}\right)\mathrm{d}t\\ &= \sum_{i=1}^n \frac{W_t^i \,\mathrm{d}W_t^i}{R_t} + \frac{n-1}{2R_t}\,\mathrm{d}t.\end{aligned}$$

Note now that $\mathrm{d}\langle R\rangle_t = (\mathrm{d}R_t)^2 = \sum_{i=1}^n \frac{(W_t^i)^2}{R_t^2}\,\mathrm{d}t = \mathrm{d}t$. Then, for $n = 2$, by Itô's Lemma again,

$$\begin{aligned}\mathrm{d}(\ln(R_t)) &= \frac{\mathrm{d}R_t}{R_t} - \frac{\mathrm{d}\langle R\rangle_t}{2R_t^2}\\ &= \frac{W_t^1\,\mathrm{d}W_t^1}{R_t^2} + \frac{W_t^2\,\mathrm{d}W_t^2}{R_t^2} + \frac{1}{2R_t}\,\mathrm{d}t - \frac{\mathrm{d}t}{2R_t^2}\\ &= \frac{W_t^1\,\mathrm{d}W_t^1 + W_t^2\,\mathrm{d}W_t^2}{R_t^2}.\end{aligned}$$

Similarly, for $n \geq 3$,

$$\begin{aligned}\mathrm{d}(R_t^{2-n}) &= (2-n)R_t^{1-n}\,\mathrm{d}R_t + \frac{1}{2}(2-n)(1-n)R_t^{-n}\,\mathrm{d}\langle R\rangle_t\\ &= (2-n)R_t^{-n}\sum_{i=1}^n W_t^i\,\mathrm{d}W_t^i + \frac{1}{2}(2-n)(n-1)R_t^{-n}\,\mathrm{d}t + \frac{1}{2}(2-n)(1-n)R_t^{-n}\,\mathrm{d}t\\ &= (2-n)R_t^{-n}\sum_{i=1}^n W_t^i\,\mathrm{d}W_t^i.\end{aligned}$$

**3.8:** Note that for $t < \infty$ we have

$$I_t^N = \int_0^t \varphi_u^N\,\mathrm{d}W_u = \begin{cases} N, & \text{if } H_{-1,N} < t, W_{H_{-1,N}} = N\\ -1, & \text{if } H_{-1,N} < t, W_{H_{-1,N}} = -1\\ W_t, & \text{otherwise.}\end{cases}$$

But it can be shown, for example by using the fact that $\limsup_t |W_t| = \infty$ almost surely, that as we let our time run to infinity, our Brownian Motion $W$ almost surely hits $-1$ or $N$, or $\mathbb{P}(W_t \in (-1, N)\ \forall t \in [0,\infty)) = 0$. So then,

$$I^N = \int_0^\infty \varphi_u^N\,\mathrm{d}W_u = \lim_{t\to\infty}\int_0^t \varphi_u^N\,\mathrm{d}W_u = \begin{cases} N, & \text{if } W_{H_{-1,N}} = N\\ -1, & \text{if } W_{H_{-1,N}} = -1.\end{cases}$$

Since, as mentioned above, $H_{-1,N} < \infty$ almost surely, we have that $\mathbb{E}\left[I^N\right] = 0$, and by continuity of Brownian paths, $W_{H_{-1,N}} \in \{-1, N\}$ almost surely. Then we can write this as

$$\begin{aligned} 0 = \mathbb{E}\left[I^N\right] &= -\mathbb{P}\left(W_{H_{-1,N}} = -1\right) + N\mathbb{P}\left(W_{H_{-1,N}} = N\right) \\ \implies \mathbb{P}\left(W_{H_{-1,N}} = N\right) &= \frac{1}{N}\mathbb{P}\left(W_{H_{-1,N}} = -1\right). \end{aligned}$$

But also, $\mathbb{P}\left(W_{H_{-1,N}} = N\right) + \mathbb{P}\left(W_{H_{-1,N}} = -1\right) = 1$, and so we can solve to find that

$$\mathbb{P}\left(W_{H_{-1,N}} = N\right) = \frac{1}{1+N} \qquad \mathbb{P}\left(W_{H_{-1,N}} = -1\right) = \frac{N}{1+N}.$$

Then, for $\varepsilon > 0$, since $W_{H_{-1}} = -1$ almost surely,

$$\lim_{N\to\infty} \mathbb{P}\left(|W_{H_{-1,N}} - W_{H_{-1}}| > \varepsilon\right) = \lim_{N\to\infty} \mathbb{P}\left(W_{H_{-1,N}} = N\right) = 0,$$

so $I^N \to I = -1$ in probability. Clearly then $\mathbb{E}[I] = -1$, and we can see that

$$\begin{aligned} \mathbb{E}\left[\left(I^N - I\right)^2\right] &= 0\mathbb{P}\left(W_{H_{-1,N}} = -1\right) + (N+1)^2\mathbb{P}\left(W_{H_{-1,N}} = N\right) \\ &= N + 1 \\ &\to \infty \quad \text{as } N \to \infty, \end{aligned}$$

so we do not have convergence in $\mathcal{L}^2(\mathbb{P})$.